The University of Azad Jammu and Kashmir

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Roll NO: 2024-SE-14

Semester:4th

Course Title: Database System

Course Code: CS-2204

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LAB 09

1. INTRODUCTION

This lab was performed to understand SQL JOIN operations using MySQL in XAMPP. Different types of joins including INNER JOIN, LEFT JOIN, RIGHT JOIN, SELF JOIN, and multi-table joins were used to retrieve related data from multiple tables in the company schema.

2. OBJECTIVE

The objective of this lab was to practice SQL JOIN operations and retrieve related records from multiple database tables.

3. Create Database

4. Company schema tables created successfully.

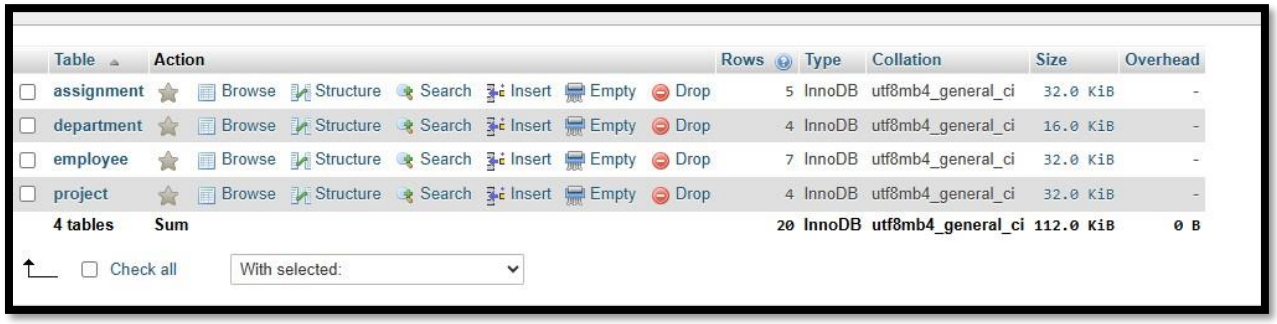

Table	Action	Rows	Type	Collation	Size	Overhead
assignment	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
department	Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	16.0 KiB	-
employee	Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_general_ci	32.0 KiB	-
project	Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	32.0 KiB	-
4 tables	Sum	20	InnoDB	utf8mb4_general_ci	112.0 KiB	0 B

Check all With selected:

Sample data inserted into all company schema tables.

```
SELECT * FROM `assignment`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	EmpID	ProjectID	WeeklyHours
Edit Copy Delete	1	1	20
Edit Copy Delete	2	3	15
Edit Copy Delete	3	2	18
Edit Copy Delete	6	1	25
Edit Copy Delete	7	4	12

Check all *With selected:* Edit Copy Delete Export

Showing rows 0 - 3 (4 total, Query took 0.0000 seconds.)

```
SELECT * FROM `department`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort

Extra options

	DeptID	DeptName	Location
Edit Copy Delete	1	Engineering	Lahore
Edit Copy Delete	2	HR	Karachi
Edit Copy Delete	3	Marketing	Islamabad
Edit Copy Delete	4	Finance	Lahore

Check all *With selected:* Edit Copy Delete Export

```
SELECT * FROM `employee`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	EmpID	EmpName	Salary	City	DeptID	ManagerID
Edit Copy Delete	1	Ali Khan	95000.00	Lahore	1	NULL
Edit Copy Delete	2	Sara Ahmed	72000.00	Karachi	2	1
Edit Copy Delete	3	Bilal Hussain	68000.00	Islamabad	3	1
Edit Copy Delete	4	Maham Noor	88000.00	Lahore	2	1
Edit Copy Delete	5	Usman Tariq	120000.00	Karachi	1	NULL
Edit Copy Delete	6	Areeba Khan	85000.00	Lahore	1	5
Edit Copy Delete	7	Daniyal Raza	76000.00	Islamabad	4	5

```
SELECT * FROM `project`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key:

Extra options

	ProjectID	ProjectName	DeptID	StartDate
Edit Copy Delete	1	Mobile App	1	2024-01-10
Edit Copy Delete	2	Website Redesign	3	2024-03-15
Edit Copy Delete	3	HR System	2	2023-11-20
Edit Copy Delete	4	Finance Tracker	NULL	2024-02-05

Check all *With selected:* Edit Copy Delete Export

5.Part B Queries

TaskB1

- This query displays each employee along with their manager's name using SELF JOIN.

```
SELECT e.EmpName AS
EmployeeName,
m.EmpName AS ManagerName
FROM Employee e
LEFT JOIN Employee m
ON e.ManagerID = m.EmpID;
```

Showing rows 0 - 6 (7 total, Query took 0.0006 seconds.)

```
-- Task B1 SELECT e.EmpName AS EmployeeName, m.EmpName AS ManagerName FROM Employee e LEFT JOIN Employee m ON e.ManagerID = m.EmpID;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

EmployeeName	ManagerName
Ali Khan	NULL
Sara Ahmed	Ali Khan
Bilal Hussain	Ali Khan
Maham Noor	Ali Khan
Usman Tariq	NULL
Areeba Khan	Usman Tariq
Daniyal Raza	Usman Tariq

Output of Task B1 showing employees and their managers.

TaskB2

- This query lists employees who earn more salary than their managers.

```
SELECT e.EmpName,
e.Salary AS EmployeeSalary,
m.EmpName AS ManagerName,
m.Salary AS ManagerSalary FROM Employee e
INNER JOIN Employee m
ON e.ManagerID = m.EmpID
WHERE e.Salary > m.Salary;
```

Output of Task B2 showing employees earning more than their managers.

TaskB3

- This query displays employees whose managers work in different departments.

```
SELECT e.EmpName,
m.EmpName AS ManagerName,
d1.DeptName AS EmployeeDepartment, d2.DeptName AS ManagerDepartment
FROM Employee e
INNER JOIN Employee m
ON e.ManagerID = m.EmpID
INNER JOIN Department d1
ON e.DeptID = d1.DeptID
INNER JOIN Department d2
ON m.DeptID = d2.DeptID
WHERE e.DeptID <> m.DeptID;
```

Showing rows 0 - 3 (4 total, Query took 0.0009 seconds.)

```
-- Task B3 SELECT e.EmpName, m.EmpName AS ManagerName, d1.DeptName AS EmployeeDepartment, d2.DeptName AS ManagerDepartment FROM Employee e INNER JOIN Employee m ON e.ManagerID = m.EmpID INNER JOIN Department d1 ON e.DeptID = d1.DeptID INNER JOIN Department d2 ON m.DeptID = d2.DeptID WHERE e.DeptID <> m.DeptID;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

EmpName	ManagerName	EmployeeDepartment	ManagerDepartment
Sara Ahmed	Ali Khan	HR	Engineering
Maham Noor	Ali Khan	HR	Engineering
Bilal Hussain	Ali Khan	Marketing	Engineering
Daniyal Raza	Usman Tariq	Finance	Engineering

Output of Task B3 showing employees and managers from different departments

TaskB4

- This query displays employees with project names and weekly working hours.

```
SELECT e.EmpName, p.ProjectName,
a.WeeklyHours FROM
Employee e
INNER JOIN Assignment a
ON e.EmpID = a.EmpID
INNER JOIN Project p
ON a.ProjectID = p.ProjectID;
```

Showing rows 0 - 4 (5 total, Query took 0.0006 seconds.)

```
-- Task B4 SELECT e.EmpName, p.ProjectName, a.WeeklyHours FROM Employee e INNER JOIN Assignment a ON e.EmpID = a.EmpID INNER JOIN Project p ON a.ProjectID = p.ProjectID;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

EmpName	ProjectName	WeeklyHours
Ali Khan	Mobile App	20
Areeba Khan	Mobile App	25
Bilal Hussain	Website Redesign	18
Sara Ahmed	HR System	15
Daniyal Raza	Finance Tracker	12

Output of Task B4 showing employees with assigned projects and hours.

TaskB5

- This query displays assignment details including employee name, project name, and department name.

```
SELECT e.EmpName,
p.ProjectName,
d.DeptName FROM
Assignment a
INNER JOIN Employee e
ON a.EmpID = e.EmpID
INNER JOIN Project p
ON a.ProjectID = p.ProjectID
LEFT JOIN Department d
ON p.DeptID = d.DeptID;
```

Showing rows 0 - 4 (5 total, Query took 0.0006 seconds.)

```
-- Task B5 SELECT e.EmpName, p.ProjectName, d.DeptName FROM Assignment a INNER JOIN Employee e ON a.EmpID = e.EmpID INNER JOIN Project p ON
a.ProjectID = p.ProjectID LEFT JOIN Department d ON p.DeptID = d.DeptID;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

EmpName	ProjectName	DeptName
Ali Khan	Mobile App	Engineering
Areeba Khan	Mobile App	Engineering
Bilal Hussain	Website Redesign	Marketing
Sara Ahmed	HR System	HR
Daniyal Raza	Finance Tracker	*NULL*

Output of Task B5 showing assignment and project department details.

TaskB6

- This query lists employees working on the Mobile App project with their weekly hours.

```
SELECT e.EmpName,
a.WeeklyHours FROM
Employee e
INNER JOIN Assignment a
ON e.EmpID = a.EmpID
INNER JOIN Project p
ON a.ProjectID = p.ProjectID
WHERE p.ProjectName =
'Mobile App';
```

Showing rows 0 - 1 (2 total, Query took 0.0007 seconds.)

```
-- Task B6 SELECT e.EmpName, a.WeeklyHours FROM Employee e INNER JOIN Assignment a ON e.EmpID = a.EmpID INNER JOIN Project p ON a.ProjectID =
p.ProjectID WHERE p.ProjectName = 'Mobile App';
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

EmpName	WeeklyHours
Ali Khan	20
Areeba Khan	25

Output of Task B6 showing employees working on the Mobile App project.

TaskB7

- This query displays Lahore employees with their assigned projects and weekly hours including employees without assignments.

```
SELECT e.EmpName, p.ProjectName,
a.WeeklyHours FROM
Employee e
LEFT JOIN Assignment a
ON e.EmpID = a.EmpID
LEFT JOIN Project p
ON a.ProjectID = p.ProjectID
WHERE e.City = 'Lahore';
```

Showing rows 0 - 2 (3 total, Query took 0.0005 seconds.)

```
-- Task B7 SELECT e.EmpName, p.ProjectName, a.WeeklyHours FROM Employee e LEFT JOIN Assignment a ON e.EmpID = a.EmpID LEFT JOIN Project p ON
a.ProjectID = p.ProjectID WHERE e.City = 'Lahore';
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

EmpName	ProjectName	WeeklyHours
Ali Khan	Mobile App	20
Maham Noor	NULL	NULL
Areeba Khan	Mobile App	25

Output of Task B7 showing Lahore employees and their projects.

TaskB8

- This query lists employees working on projects managed by departments different from their own departments.

```
SELECT DISTINCT e.EmpName
FROM Employee e
INNER JOIN Assignment a
ON e.EmpID = a.EmpID
INNER JOIN Project p
ON a.ProjectID = p.ProjectID
WHERE e.DeptID <> p.DeptID;
```

Output of Task B8 showing employees working across different departments

TaskB9

- This query displays departments and projects started in the year 2024 including departments without projects.

```
SELECT d.DeptName,
p.ProjectName FROM
Department d
LEFT JOIN Project p
ON d.DeptID = p.DeptID
AND YEAR(p.StartDate) = 2024;
```

Showing rows 0 - 3 (4 total, Query took 0.0005 seconds.)

```
-- Task B9 SELECT d.DeptName, p.ProjectName FROM Department d LEFT JOIN Project p ON d.DeptID = p.DeptID AND YEAR(p.StartDate) = 2024;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

DeptName	ProjectName
Engineering	Mobile App
Marketing	Website Redesign
HR	*NULL*
Finance	*NULL*

Output of Task B9 showing departments with projects started in 2024.

TaskB10

- This query displays every employee with total weekly working hours across all projects including employees with zero hours.

```
SELECT e.EmpName,
IFNULL(SUM(a.WeeklyHours),0)
AS TotalHours
FROM Employee e
LEFT JOIN Assignment a
ON e.EmpID = a.EmpID
GROUP BY e.EmpName;
```

Showing rows 0 - 6 (7 total, Query took 0.0011 seconds.)

```
-- Task B10 SELECT e.EmpName, IFNULL(SUM(a.WeeklyHours),0) AS TotalHours FROM Employee e LEFT JOIN Assignment a ON e.EmpID = a.EmpID GROUP BY e.EmpName;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

EmpName	TotalHours
Ali Khan	20
Areeba Khan	25
Bilal Hussain	18
Daniyal Raza	12
Maham Noor	0
Sara Ahmed	15
Usman Tariq	0

Output of Task B10 showing total weekly hours for each employee.

6. Conclusion

In this lab, different SQL JOIN operations were performed successfully using MySQL in XAMPP. INNER JOIN, LEFT JOIN, RIGHT JOIN, SELF JOIN, and multi-table joins helped in retrieving related information from multiple tables. The lab improved understanding of relational database concepts and SQL query writing.

*****THE END*****